Maitrey Neeraj Patankar

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.08
Holy Cross Senior Secondary School	Class XII (CBSE)	2017	71.80%
Holy Cross Senior Secondary School	Class X (CBSE)	2014	8.60%

WORK EXPERIENCE

Infosys | SmartMaint - AI-Driven Predictive Maintenance System | *ServiceNow* | *Javascript* | *Flow Designer* |

- Developed an ML predictive system achieving 92% recall for equipment failure risk to enable proactive interventions.
- Integrated models with ServiceNow to automate alerts and work orders, reducing manual processing steps by 40%.

Infosys | ServiceOps AI - Intelligent ITSM Incident Management | *ServiceNow* | *NLP* |*REST APIs* |

- Developed an **NLP-based incident classification and intelligent-routing engine** integrated with ServiceNow ITSM, achieving **91% classification accuracy** and reducing manual ticket-triage effort by **~35%**.
- Built an **SLA-breach prediction and automated escalation pipeline**, identifying **~85% of at-risk incidents** before simulated SLA violations and enabling proactive service-management intervention.

Infosys | ServiceAssist - Enterprise GenAI/RAG Service Desk | *ServiceNow* | *LLM* | *RAG* | *Vector Database* |

- **Built a RAG-powered ServiceNow assistant** with **~90% retrieval success** for enterprise support queries.
- **Automated ServiceNow request creation and routing**, cutting manual processing by **~45%**.

RESEARCH EXPERIENCE

AI/ML for Carbon Capture, Utilization & Storage (CCUS) | Review Research |

- Developed a **systematic AI-assisted literature-screening methodology and Python toolkit**, reducing **15,000+ records to ~200 relevant studies** through relevance screening, classification and structured information extraction.
- Conducting **manual validation and evidence synthesis** to assess AI/ML methodologies, engineering applicability, model validation and deployment readiness across carbon capture, utilisation and storage.

M.Tech Thesis - Physics-Informed Neural Networks for Ammonia Combustion |

- Developing **PINNs integrating CFD data and governing combustion physics** for predictive modelling of ammonia combustion and optimised fuel-blending strategies to address low flame-speed limitations.

RELEVANT COURSEWORK

Sustainability Oriented - Concepts of Sustainability, Sustainable Building Materials, CCUS, Photovoltaics and Solar Economy, Sustainable Energy Technologies, Chemistry for Sustainable Energy, SWOT Analysis, Supply Chain and Circular Economy

Climate Change Oriented - Entrepreneurship in Climate Change, Climate Modeling, RS, GIS, Climate Data Analysis

AI/ML Tech oriented - Machine Learning, Deep Learning, NLP, Statistical Analysis, Physics Informed Modeling, Reinforcement Learning, Computer Vision, Data Science and Analytics, Optimization Techniques.

PROJECTS

Wind Turbine SCADA Multi-Output Regression using Artificial Neural Networks | PyTorch | Model Evaluation

- **Optimized a PyTorch multi-output ANN** on 26K+ SCADA rows to simultaneously predict wind turbine power and

rotor RPM.
- Achieved **R2 = 0.943 for power** and **R2 = 0.970 for rotor RPM** through ANN-based predictive modelling and model optimization.

Multi-Objective Energy Allocation Optimisation for Mumbai | Python, MILP, Regression |

- Developed a **multi-objective model** across 8 energy sources, projecting a **96% coal phase-down** and **7,160 MW solar growth**.
- Integrated **demand forecasting** with a **40:60 eco-environmental optimisation**, mapping a **569% hydro-capacity expansion pathway**.

ANN-Assisted Multi-Objective Rice Production Optimisation | TensorFlow, NSGA-III |

- Built an **iterative ANN surrogate (128-64-32)** on a **35+ dimensional pan-India dataset** for production and pesticide prediction.
- Coupled ANN with **NSGA-II** to generate **Pareto-optimal production-pesticide trade-offs**.

Sustainable Multi-Objective Supply Chain Network Optimisation | PuLP, MILP |

- Modeled a **5-country MILP supply-chain network** to optimize capacity, freight, and service constraints under strict **CO2 caps**.
- Developed an **epsilon-constraint cost-carbon Pareto framework** identifying a **38.6% CO2 reduction** at a **10.0% cost premium**.

Domain-Robust Deep Learning for Steel Microstructure Bainite-Martensite Classification

- **Built a domain-robust CV pipeline** for classification across **5,293 micrographs** using CNN, ResNet18, ViT-B/16 and Swin-T.
- **Exposed a 29-point generalization gap** in ResNet18 Macro-F1 (**0.961 to 0.670**) using leakage-safe specimen, microscope and magnification holdouts.
- **Boosted unseen-specimen Macro-F1 to 0.807** with acquisition-aware Swin-T, achieving a **+13.7-point gain over ResNet18**.

SKILLS

Sustainability & Climate: CCUS, Carbon/ESG Accounting, LCA, Energy Systems Modelling, Climate Modelling, Sustainable Supply Chains, Renewable Energy Systems, AI for Climate & Sustainability,

Programming & Data: Python, NumPy, Pandas, SciPy, Matplotlib, Jupyter, Git/GitHub

Machine Learning: Scikit-learn, Regression, ANN, Predictive Modelling, Surrogate Modelling, Model Evaluation

Deep Learning & GenAI: TensorFlow, PyTorch, NLP, Transformers, RAG, PINNs

Optimisation & Analytics: NSGA-II/III, MILP, Differential Evolution, Sobol Sensitivity Analysis, Multi-Objective Optimisation

POSITIONS OF RESPONSIBILITY

Organising Committee Member, Savinya 2026, IIT Hyderabad

- Oversaw **100+ members** and managed end-to-end event operations, venue, hospitality, logistics and on-ground execution.

- Led corporate and sponsorship outreach, pitching partnership proposals of up to **Rs. 2 lakh** to prospective companies.

Placement Coordinator, Office of Career Services (OCS), IIT Hyderabad

- Led outreach to **50+ companies**, building relations and expanding career opportunities for an **80+ student cohort**.

- Organised **30+ corporate engagements**, recruiter interactions, career talks and career-fair outreach with industries.

SCHOLASTIC ACHIEVEMENTS

Awarded the **Ministry of Education (MoE) Postgraduate Scholarship** for pursuing M.Tech at IIT Hyderabad.

CERTIFICATIONS

- **Global Consumer Intelligence (GCI) Data Science Program** -- The University of Tokyo, Matsuo-Iwasawa Laboratory
- **Machine Learning Specialization** -- DeepLearning.AI & Stanford Online, Andrew Ng
- **Deep Learning Specialization** -- DeepLearning.AI, Andrew Ng

- **Climate Change AI (CCAI) Virtual Summer School 2026** - Training covered AI for climate science, buildings & cities, agriculture & food security, water resources, policy/economics and responsible AI.